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Diagnosing neural network training

1. Data and settings

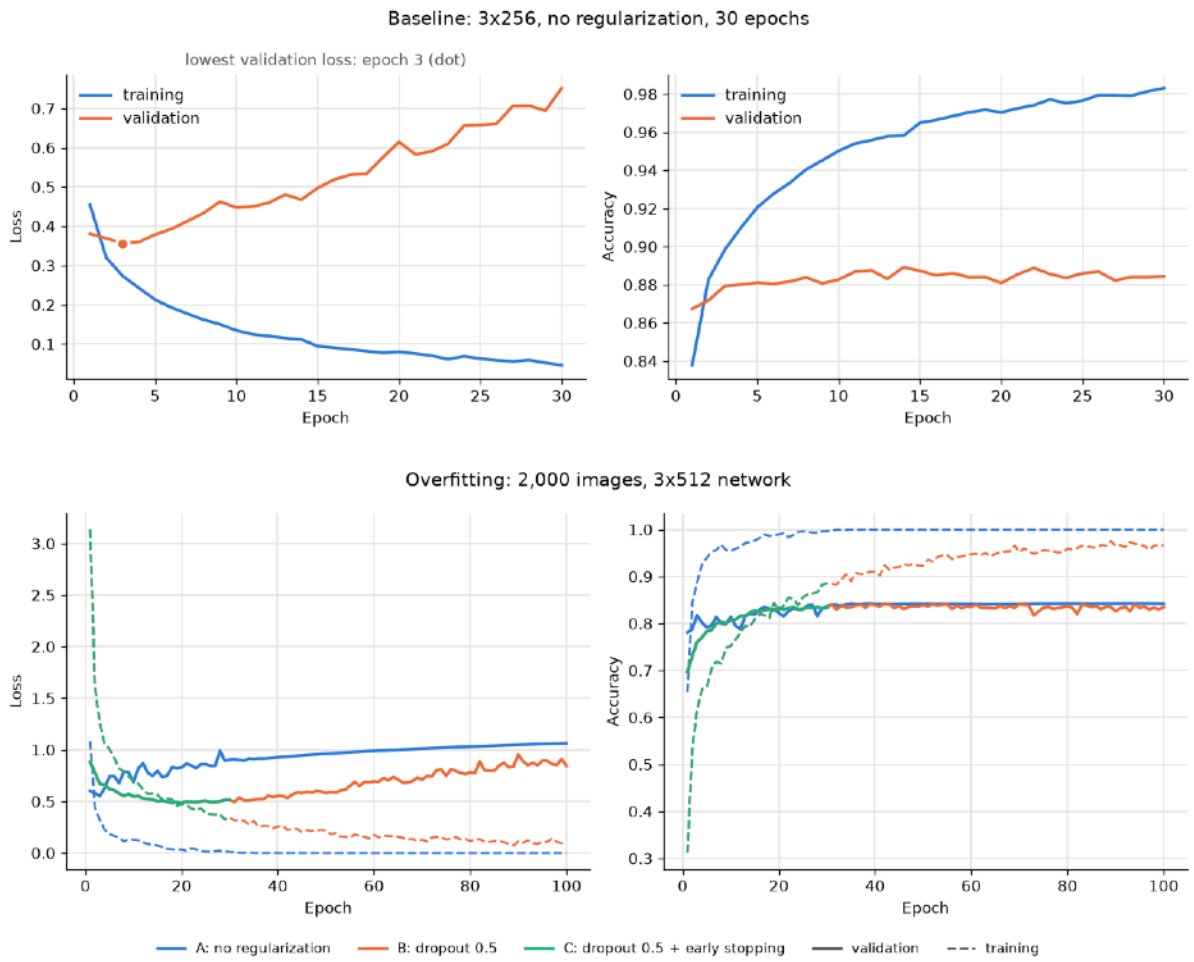


Data	Fashion-MNIST: 70,000 grayscale images, 28×28 pixels (0–255), 10 balanced classes; each image flattened to 784 features.
Split	Training 50,000 / validation 10,000 (1000 per class) / test 10,000
Network (baseline)	784 -> 256 ReLU -> 256 ReLU -> 256 ReLU -> 10 softmax; 335,114 parameters

Runs

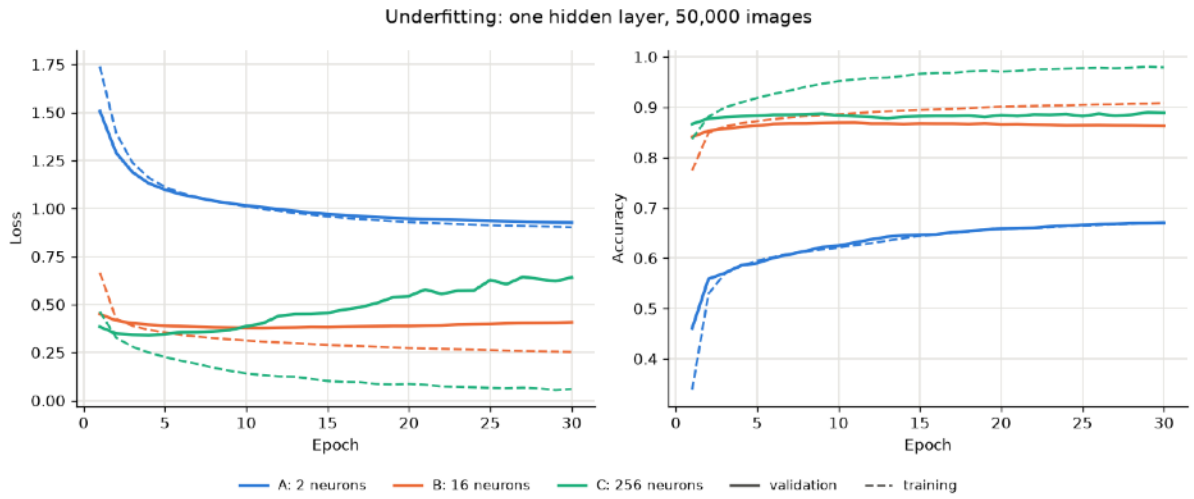
Run	Hidden layers	Training images	Dropout	Optimizer	Epochs	Early stopping
Baseline	3×256	50,000	0	Adam	30	no
Overfitting A	3×512	2,000	0	Adam	100	no
Overfitting B	3×512	2,000	0.5	Adam	100	no
Overfitting C	3×512	2,000	0.5	Adam	≤ 100 (ran 30)	yes
Underfitting A / B / C	1×2 / 1×16 / 1×256	50,000	0	Adam	30	no
Experiment 1: dropout	3×256	50,000	0 / 0.3 / 0.5 / 0.7	Adam	30	no
Experiment 2: optimizer	3×256	50,000	0, then 0.3	SGD / SGD + mom. / Adam / AdamW	30, then ≤ 100	no, then yes
Final model	3×256	50,000	0.3	Adam	≤ 100 (ran 31)	yes

2. Baseline and problem 1: overfitting



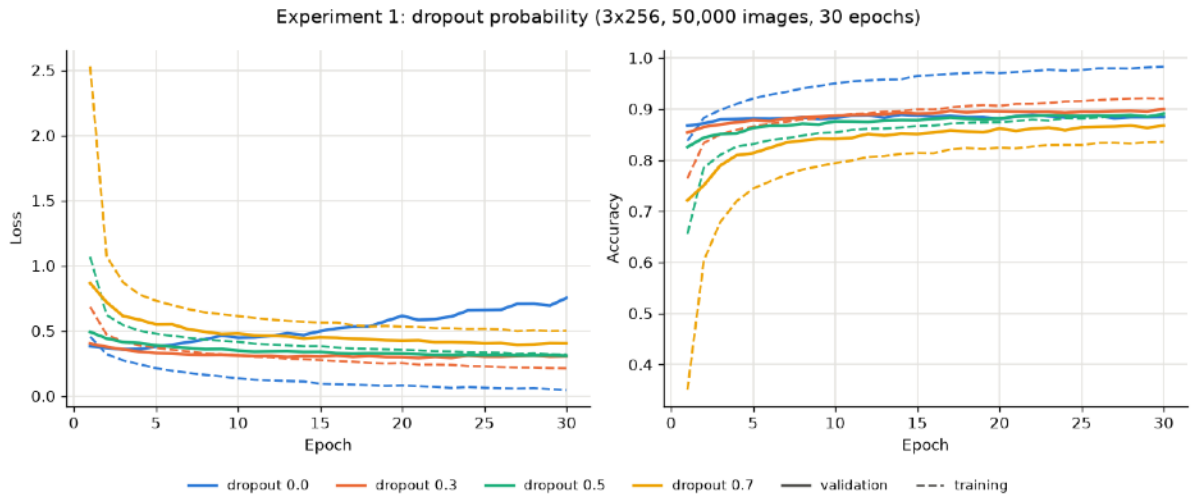
Run	Epochs run	Best epoch	Lowest val. loss	Val. loss	Training acc.	Val. acc.	Gap
Baseline (50,000 images, 3×256)	30	3	0.3566	0.7517	97.29%	88.44%	8.8
A: no regularization	100	3	0.5526	1.0646	100.00%	84.25%	15.8
B: dropout 0.5	100	20	0.4832	0.8452	99.80%	83.50%	16.3
C: dropout 0.5 + early stopping	30	20	0.4832	0.4832	91.65%	83.32%	8.3

3. Problem 2: underfitting



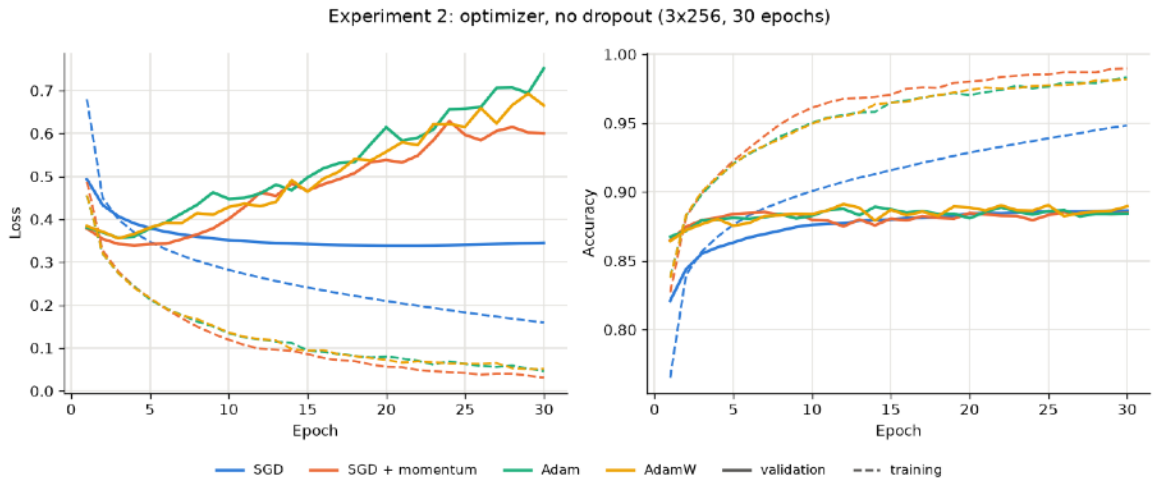
Run (one hidden layer)	Parameters	Best epoch	Lowest val. loss	Val. loss	Training acc.	Val. acc.	Gap (points)
A: 2 neurons	1,600	30	0.9266	0.9266	67.45%	66.97%	0.5
B: 16 neurons	12,730	11	0.3781	0.4084	90.36%	86.25%	4.1
C: 256 neurons	203,530	4	0.3414	0.6408	97.36%	88.84%	8.5

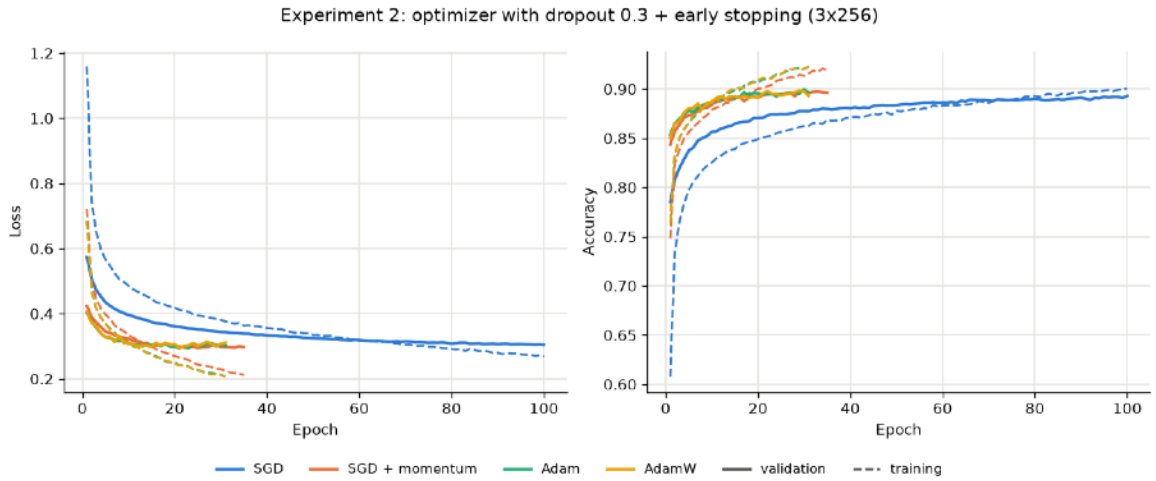
4. Experiment 1: dropout probability



Run	Epochs run	Best epoch	Lowest val. loss	Val. loss	Training acc.	Val. acc.	Gap (points)
dropout 0.0	30	3	0.3566	0.7517	97.29%	88.44%	8.8
dropout 0.3	30	21	0.2929	0.3037	94.71%	89.97%	4.7
dropout 0.5	30	30	0.3074	0.3074	91.75%	89.08%	2.7
dropout 0.7	30	27	0.3933	0.4032	88.07%	86.76%	1.3

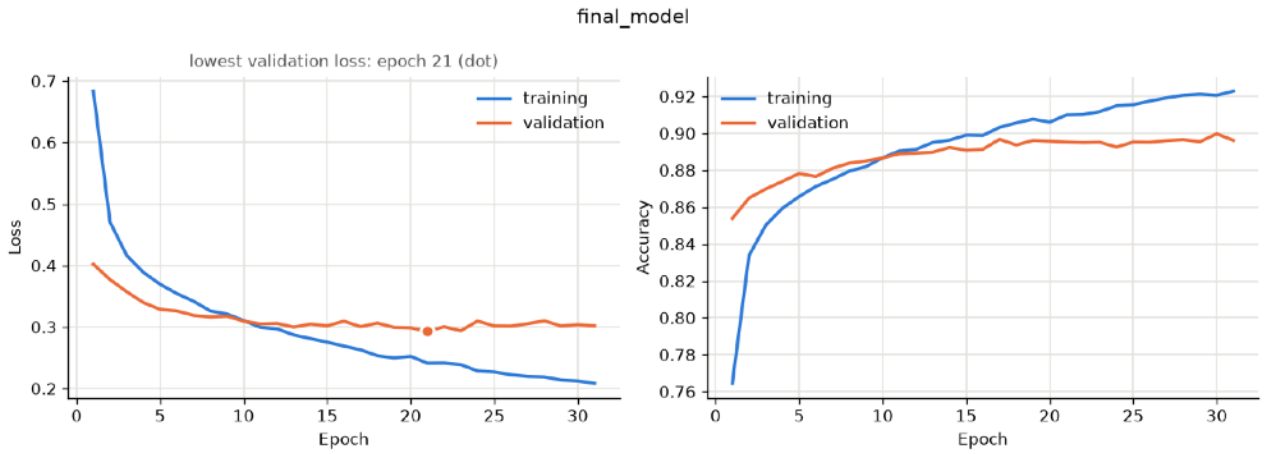
5. Experiment 2: optimizer





Optimizer	Settings	No dropout: lowest val. loss (epoch)	No dropout: val. loss at 30	Dropout + ES: epochs	Dropout + ES: lowest val. loss (epoch)	Dropout + ES: val. acc.	Dropout + ES: gap (points)
SGD	lr 0.01	0.3389 (22)	0.3453	100	0.3039 (100)	89.30%	3.5
SGD + momentum	lr 0.01, momentum 0.9	0.3393 (4)	0.5998	35	0.2941 (25)	89.46%	4.2
Adam	lr 0.001	0.3566 (3)	0.7517	31	0.2929 (21)	89.52%	3.8
AdamW	lr 0.001, weight decay 0.004	0.3557 (3)	0.6654	31	0.2936 (21)	89.53%	3.9

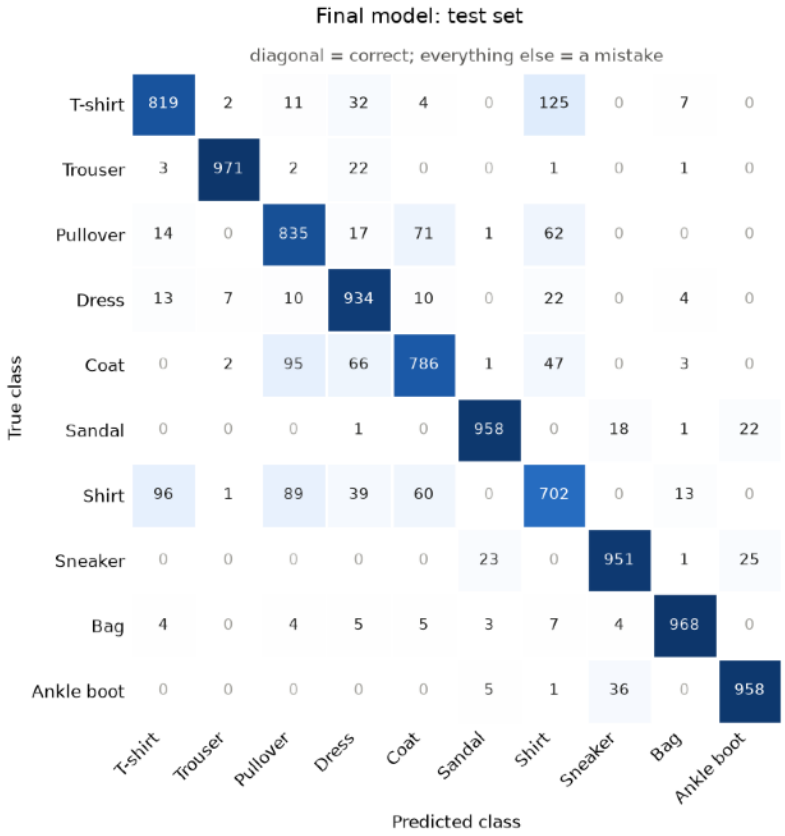
6. Final model and test results



The final configuration combines the best results of sections 2–5 and was chosen on validation data only. The test set was evaluated once, after all decisions had been made.

Network	784 -> 256 ReLU -> dropout 0.3 -> 256 ReLU -> dropout 0.3 -> 256 ReLU -> dropout 0.3 -> 10 softmax
Training	Adam, learning rate 0.001; batch size 128; max. 100 epochs; EarlyStopping (val_loss, patience 10, restore_best_weights) – stopped after 31 epochs, weights of epoch 21 restored.
Validation	loss 0.2929; accuracy 89.52%; training accuracy 93.35%; gap 3.8 points.

Test set



Per class

Class	Precision	Recall	F1
T-shirt	0.863	0.819	0.840
Trouser	0.988	0.971	0.979
Pullover	0.798	0.835	0.816
Dress	0.837	0.934	0.883
Coat	0.840	0.786	0.812
Sandal	0.967	0.958	0.962
Shirt	0.726	0.702	0.714
Sneaker	0.943	0.951	0.947
Bag	0.970	0.968	0.969
Ankle boot	0.953	0.958	0.956

Most confident mistakes on the test set (p = probability given to the wrong class)

